

SANGJUN (SAMUEL) Ko

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EDUCATION

University of Illinois Urbana-Champaign

Master of Science in Statistics

Bachelor of Science in Statistics (CGPA: 3.92/4.00)

Champaign, Illinois

August 2025 - May 2027 (Expected)

January 2023 - May 2025

- Mathematics Minor
- Coursework: Data Science Programming Methods(STAT447), Advanced Data Analysis(STAT448), Statistical Modeling II(STAT426), Statistical Data Management(STAT440), Statistics and Probability II(STAT410), Elementary Real Analysis(MATH444), Linear Algebra(MATH257)

University of Michigan-Shanghai Jiao Tong University Joint Institute

Shanghai, China

Bachelor of Engineering in Mechanical Engineering (CGPA: 3.50/4.00)

September 2019 - December 2020

- First Class Scholarship

WORK EXPERIENCE

STAT107 Data Science Discovery

Undergraduate Course Assistant

Champaign, Illinois

August 2023 - Present

- Facilitate learning for over 100 students by assisting students in grasping fundamental data science concepts by conducting interactive sessions, and providing clarifications on course materials
- Manage Git for learning platform modifications, benefiting 1000+ students; create, edit, and proofread course materials
- Collaborate with a team of 100+ staff and 2 professors in weekly meetings

STAT443 Professional Statistics

Undergraduate Course Assistant

Champaign, Illinois

January 2025 - Present

- Grade assignments and provide feedback to 20+ students on statistical analysis and consulting using Canvas LMS
- Evaluate 5+ student presentations to ensure alignment with course expectations and grading criteria
- Act as a client in mock consulting projects, providing insights and posing real-world challenges

Applied Technologies for Learning in Arts & Sciences

AI/ML Intern Team Leader

Champaign, Illinois

January 2024 - December 2024

- Lead a team of 5 interns.
- Develop predictive machine learning models with large data sets(100,000+ entries) using Python
- Utilize advanced techniques: XGBoost, Random Forests, Regression, Neural Networks, BERT, Transformers
- Collaborate closely with 3 coordinators in weekly meetings to align project goals and track progress

HappySneeze

Market Research Intern

Remote

August 2024 - September 2024

- Conducted market research and consumer trend analysis to enhance understanding of menopausal women's health needs
- Led outreach and interviews with target demographics to gather insights on pelvic floor health
- Synthesized findings into actionable strategies and presented recommendations to stakeholders

KM Internacional S.A.

Intern

Managua, Nicaragua

May 2023 - August 2023

- Oversaw comprehensive inventory management and streamlined administrative tasks for a small textile company
- Lead a team of three workers to achieve a 98% accuracy rate in record-keeping and a 15% reduction in operational costs

Korean Augmentation to the United States Army Program

Soldier

Gyeonggi-do, South Korea

Jun 2021 - Dec 2022

- Demonstrated exceptional service in the "Korean Augmentation to the United States Army" program, completing a rigorous 18-month military assignment
- Maintained and operated an M1 Abrams Tank, showcasing expertise as a Tank Infantry Soldier within the 2nd Squadron, 13th Cavalry Regiment
- Streamlined inventory management operations as an S-4 Soldier within the 176th Financial Management Support Unit, ensuring accurate and efficient financial processes

Istmo Textiles S.A.

Intern

Masaya, Nicaragua

Feb 2021 - Mar 2021

- Worked as an intern in a Textiles Corporation mainly learning about the social responsibility of corporations; human rights, labor laws, environmental laws, and industrial safety laws
- Studied the import and export of clothing and textiles from Central America to North America

- Attended corporate social responsibility activities scheduled by the corporation aimed at supporting impoverished schools and churches in Masaya, Nicaragua, by providing school supplies and food

PROJECTS/AWARDS

- Graduation Prediction Model** September 2024 - December 2024
- Increased AUC score from 0.65 to 0.82 by reducing noise in the dataset through feature engineering and outlier removal across 50,000+ student records
 - Built an interactive Streamlit interface with 12 key predictors, enabling real-time graduation predictions for university administrators
- 3rd Place, 2024 COVID Information Commons Student Paper Challenge** September 2024
- Analyzed the socioeconomic and mental health impacts of COVID-19 in Mexico, identifying that 42% of females and 40% of young adults faced job loss
 - Proposed data-driven interventions to support vulnerable groups through geospatial and statistical analysis
- Paraphrase Detection** July 2024
- Achieved 90%+ accuracy in paraphrase detection using BERT, RoBERTa, and SBERT models on the MRPC dataset
 - Developed a Gradio interface for real-time paraphrase detection, facilitating easier evaluation of textual similarities for academic and business use cases
- International Student Forecasting** April 2024
- Processed and analyzed large-scale student enrollment data (over 100,000 entries) from UIUC to identify trends and patterns
 - Suggested resource allocation optimization by forecasting international student numbers using linear regression and deterministic processes
- Titanic Machine Learning Model** February 2024
- Developed a predictive model with an 80% accuracy using Python to determine factors influencing survival rates in the Titanic disaster using passenger data
 - Identified sex and passenger class as significant survival factors through feature importance analysis

SKILLS

Languages:	English (Fluent), Spanish (Native), Korean (Native), French (Basic), Chinese (Basic)
Computer:	Python (Advanced): Pandas, NumPy, Scikit-learn, TensorFlow, NLTK, Matplotlib, Seaborn, BERT, Random Forest, XGBoost, Gradio R (Intermediate): Tidyverse, ggplot2, dplyr, caret Git (Intermediate), SAS (Intermediate), Tableau (Basic), SQL (Basic), Office (Advanced)
Certificates:	Data Science DISCOVERY, Data Science Certificate, Coursera Business Analytics Certificate

COMMUNITY SERVICE AND EXTRACURRICULAR ACTIVITIES

- Consultant in Illinois Business Consulting** September 2024 - December 2024
- Collaborating with interdisciplinary teams to deliver strategic solutions for real-world business challenges, including market analysis, operational optimization, and growth strategy, for clients ranging from Fortune 100 companies to nonprofits
- Treasurer in Performance UIUC Touch** February 2023 - May 2024
- Strategically plan and execute 10 fundraising initiatives to support team activities, costumes, and travel expenses, resulting in a 30% increase in available funds compared to the previous year
 - Contribute to the artistic success of the team by actively participating in weekly practices, refining dance routines, and honing performance skills.
 - Showcase talent by performing at renowned events elevating the visibility of Performance UIUC Touch within the dance community and attracting 100+ new followers on social media platforms
- Translator for the US Army** June 2021-December 2022
- Served as a translator for over 2,000 US soldiers and Korean soldiers during joint USFK training exercises
 - Assisted newly stationed US soldiers in navigating South Korea, resulting in a 95% satisfaction rate among newly stationed soldiers regarding their acclimation process
- President of Asian Culture Club** Nov 2018-Mar 2019
- Lead a group of 5 students in a club that aimed to encourage a multicultural environment in a Latin American high school
 - Arranged a meeting every week and prepared an activity every month
- Volunteer in Habitat For Humanity** Sep 2016-Jun 2017
- Volunteer in a school club with the main purpose of providing general maintenance and basic necessities to the poor regions in Nicaragua.

- Participated in construction of 15 homes and hygienic systems

Teacher at Nanum School

Aug 2016-Dec 2016

- Part of a group of teachers in a local school organized by our Korean Presbyterian Church
- Taught English and Math to children of ages 5-12 in Masaya, Nicaragua